

Probabilistic Human Mesh Recovery for Accurate Shape Modelling

1 Setup

Recover accurate body shape under clothing from multiple photos of the same person, where the pose of the person varies.

1.1 MHR

We use parametric body models such as SMPL [3] and MHR [1]. Below we use MHR as the example. MHR models the human with

$$M = \mathcal{M}(\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{s}) \quad (1)$$

where $\boldsymbol{\theta}$ are pose parameters, $\boldsymbol{\beta}$ shape, and $\mathbf{s}$ scale or bone length parameters, $\mathcal{M}$ is the MHR model, and M is the human mesh. We denote the set of parameters as $\Theta = \{\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{s}\}$. Further we overload $\boldsymbol{\beta}$ to often refer to both shape and scale parameters.

1.2 Model

Given images of a person $\{I_i\}_{i=1}^N$, we wish to recover the shape of the human, represented as the low dimensional shape parameters $\boldsymbol{\beta}$ of a parametric body model.

Sam3DBody [5] is the SoTA single-view human mesh recovery model:

$$\Theta = \text{SAM3DBody}(I) \quad (2)$$

However, single view human mesh recovery is an ill-posed problem, where multiple 3D humans could explain the 2D image. Mathematically, ‘explaining the 2D image’ means minimising the reprojection error:

$$\mathcal{L} = \|\Pi(\mathcal{J}M, \mathbf{c}) - J\|_2^2 \quad (3)$$

where Π is camera projection, $\mathcal{J}$ is a pretrained joint regressor (or some other keypoints), and J are ground truth 2D joints or keypoints.

Thus, we wish to derive some method to get a more accurate body shape from multiple photos of the same human, in varying poses.

$$(\{\boldsymbol{\theta}_i\}_{i=1}^N, \boldsymbol{\beta}) = f(\{I_i\}_{i=1}^N) \quad (4)$$

Here, the poses vary, but the shape is the same across the images.

Since we do not want to train a large model due to data or compute constraints, we try to derive some method of merging the predictions from multiple images, while keeping the original large model frozen. One way is to model the probability distribution over model parameters. Given an image I_i , we train a model to predict

$$p(\Theta|I_i) = p(\boldsymbol{\theta}, \boldsymbol{\beta}, \mathbf{s}|I_i) \quad (5)$$

And to extract the final body shape, we evaluate the mode or sample from the joint posterior

$$\boldsymbol{\beta}^* = \arg \max_{\boldsymbol{\beta}} p(\boldsymbol{\beta}|\{I_i\}_{i=1}^N) \quad (6)$$

Probabilistic modelling enables us to:

- Generate samples conditioned on a single image
- Merge shape predictions from different views

Below we discuss some approaches for this specific task.

2 Modelling Pose and Shape Parameter Distributions

2.1 Diagonal Gaussian Distribution

The simplest approach is to place Gaussians over model parameters. Refer to Akash’s paper for details [4]. Briefly,

$$p(\boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{I}), \Sigma_{\boldsymbol{\theta}}(\mathbf{I})) \quad (7)$$

$$p(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\beta}}(\mathbf{I}), \Sigma_{\boldsymbol{\beta}}(\mathbf{I})) \quad (8)$$

where covariance matrices $\Sigma = \text{diag}(\boldsymbol{\sigma}^2)$ are diagonal. Probabilistic combination of the shape parameters can be done by product of Gaussians

$$p(\boldsymbol{\beta} | \{I_i\}_{i=1}^N) \propto p(\{I_i\}_{i=1}^N | \boldsymbol{\beta}) p(\boldsymbol{\beta}) \quad (9)$$

$$= p(\boldsymbol{\beta}) \prod_{i=1}^N p(I_i | \boldsymbol{\beta}) \quad (10)$$

which can be done in closed form.

Conditional independence of images. Images $(\mathbf{I}_i, \mathbf{I}_j)$ are not independent, since they capture the same person. However, when conditioned on the body shape $\boldsymbol{\beta}$, the images become conditionally independent, since there are no constraint beyond the same body shape, e.g. poses are free to vary. That is $p(\mathbf{I}_i, \mathbf{I}_j | \boldsymbol{\beta}) = p(\mathbf{I}_i | \boldsymbol{\beta}) p(\mathbf{I}_j | \boldsymbol{\beta})$.

However, we can prove that this simple modelling cannot satisfy 2D consistency. In Figure 1 we plot a motivating example. Here, we abstract away the full kinematic chain and focus on two connected bone segments, this could for example be the upper and lower arms, connected by the elbow joint. The ambiguity originates from depth ambiguity, where multiple configurations of parameters can achieve 2D consistency on the image.

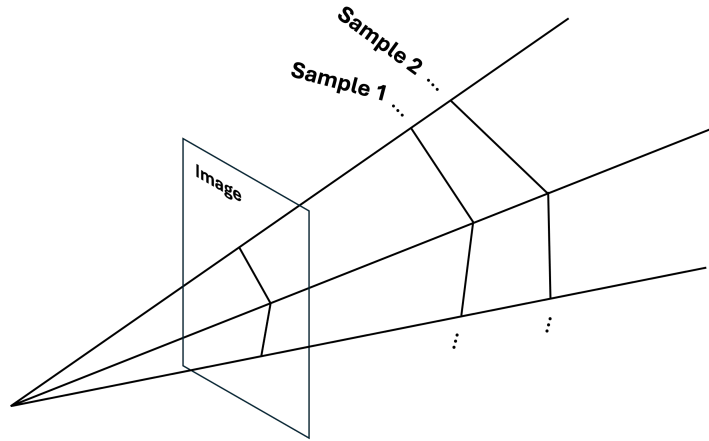


Figure 1: Samples that satisfy 2D consistency.

We know that plausible body samples should all satisfy 2D image consistency when projected onto the image. Thus, we should learn the conditional probability distribution that generates such samples conditioned on the input 2D evidence.

We learn the probability distribution in axis-angle space $\mathbb{R}^3$, which is isomorphic to $\mathfrak{so}(3) \in \mathbb{R}^{3 \times 3}$. Elements of $\mathfrak{so}(3)$ can be pushed onto $SO(3)$ using the exponential map $\exp(\cdot)$, which is the Rodrigues formula

$$\exp(\hat{\mathbf{v}}) = I + (\sin \theta)\hat{\mathbf{u}} + (1 - \cos \theta)\hat{\mathbf{u}}^2 \quad (11)$$

where $\hat{\mathbf{v}} \in \mathfrak{so}(3)$ represents an axis-angle vector $\mathbf{v}$ projected onto the Lie algebra using the hat operator.

The parameter manifold that satisfies 2D consistency in axis-angle space is complex, due to the following:

- The perspective projection is non-linear
- The kinematic chain propagates errors
- The exponential map is non-linear

Thus, we can plot an abstraction of the axis-angle manifold in Figure 2.

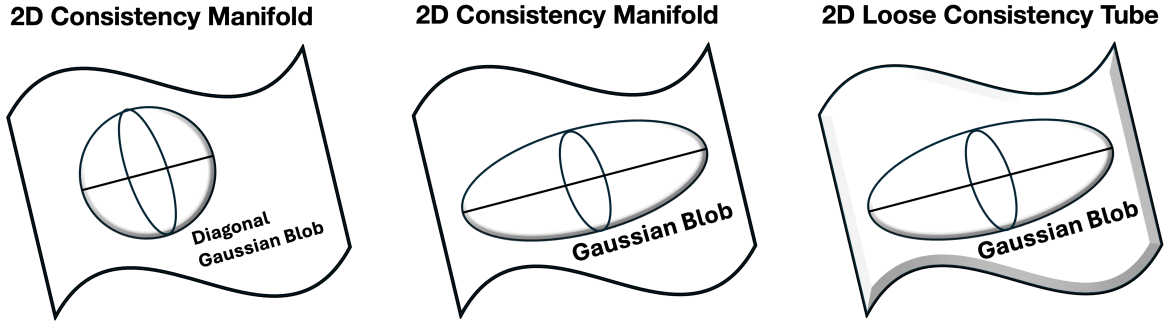


Figure 2: The rotation manifold that satisfies 2D consistency.

The left figure plots the parameter manifold on which elements satisfy the 2D consistency constraint perfectly. A diagonal Gaussian distribution cannot guarantee consistency. In fact, a great majority samples will fall outside the manifold.

2.2 Full-Covariance Per-Joint Gaussian Distribution

A simple improvement in the middle figure would be to model the full covariance matrix of the rotation. This is better since the model can capture the rough shape of the manifold, albeit coarsely.

As a side note, the manifold that satisfies strict consistency is not dimension-preserving, since for strict 2D alignment, the individual rotations are manifolds on $SO(2)$. Thus, no dimension-preserving probability distributions could fully model this manifold. In practice, 2D consistency is enforced via a loss function on joints and/or keypoints during model training. We do not and should not enforce strict consistency, since there are errors and ambiguities in the locations of the joints themselves. In this case, the consistency manifold becomes a tube, which is of the same dimension as the full rotation space.

We upgrade the model to predict the Cholesky factor L for each joint rotation axis-angle vector. We can sample from this using the reparameterisation trick

$$\Sigma = LL^T \quad (12)$$

$$\epsilon \sim \mathcal{N}(\mathbf{0}, I) \quad (13)$$

$$\boldsymbol{\theta} = \boldsymbol{\mu} + \epsilon\Sigma \quad (14)$$

This is an improvement, but still does not consider the kinematic chain, and cannot fully model the shape of the 2D consistency tubes.

Now, in theory, a normalising flow distribution can perfectly capture this, provided that the flow is flexible enough.

2.3 Normalising Flow Models for Quantifying HMR Uncertainty

We start by discussing the inference pipeline which motivates the autoregressive factorisation of our model architecture.

2.3.1 Inference

Our final goal is to learn the joint posterior $p(\beta|\{\mathbf{I}_i\}_{i=1}^N)$ of the body shape parameters $\beta = (\beta, \mathbf{s})$, here we have overloaded β to represent both the shape and bone scale parameters in the MHR model. For the same human, these parameters are fixed across different images. We wish to either:

- sample from the joint posterior $\{\beta\} \sim p(\beta|\{\mathbf{I}_i\}_{i=1}^N)$
- find the mode of it $\beta^* = \arg \max_{\beta} p(\beta|\{\mathbf{I}_i\}_{i=1}^N)$

Sampling. To sample from the marginal joint posterior would require importance sampling. For brevity, below we consider the case of two images. We can write the joint posterior as follows:

$$p(\beta|\mathbf{I}_i, \mathbf{I}_j) \propto p(\mathbf{I}_i, \mathbf{I}_j|\beta)p(\beta) \quad (15)$$

$$= p(\mathbf{I}_i|\beta)p(\mathbf{I}_j|\beta)p(\beta) \quad (16)$$

$$\propto \frac{p(\beta|\mathbf{I}_i)p(\beta|\mathbf{I}_j)}{p(\beta)} \quad (17)$$

where Eq. 16 assumes that the images are independent once conditioned on shape, which we have justified before. That is, to sample from the marginal joint posterior, we require samples and their probabilities of the conditional marginal distribution of the shape parameters $p(\beta_i|\mathbf{I}_i)$.

Once we have samples from each conditional distribution, we use importance sampling to sample from the joint posterior. Assume a (bad) uniform prior, choose a flow $p(\beta|\mathbf{I}_i)$ as the proposal $q(x)$. Compute the unnormalised importance weights for each sample

$$w_i = \frac{\text{target}(\beta_i)}{\text{proposal}(\beta_i)} = \frac{\frac{p(\beta|\mathbf{I}_i)p(\beta|\mathbf{I}_j)}{p(\beta)}}{p(\beta|\mathbf{I}_i)} = \frac{p(\beta|\mathbf{I}_j)}{p(\beta_i)} \quad (18)$$

which is simply the likelihood of the sample in the other flow. Given the unnormalised importance weights, normalise them and resample to obtain samples distributed according to the joint posterior. With proper training weight collapse should not happen since the mode predictions are close.

Mode. For different flavours of normalising flow models, the mode of the posterior distribution might not correspond to the mode of the base distribution

$$\theta^* = \arg \max_{\theta} p(\theta|I) \neq f(\mathbf{0}, \mathbf{c}) \quad (19)$$

However, for the Glow model, this does become equality. This is used by ProHMR [2]. If not, then one needs to optimise for the mode which is not desirable. Let's assume that this holds. Then, mode of the joint distribution is given by

$$\beta^* = \arg \max_{\beta} p(\beta|\mathbf{I}_i, \mathbf{I}_j) \quad (20)$$

$$= \arg \max_{\beta} p(\beta|\mathbf{I}_i)p(\beta|\mathbf{I}_j) \quad (21)$$

which again is not analytical.

2.3.2 Model

Since we would like to keep the large pre-trained model fixed, we cannot directly learn a normalising flow distribution on the model parameters Θ themselves. This would render the original model predictions obsolete. Instead, we learn the residual $p(\Delta\Theta|I, \hat{\Theta})$ conditioned on the model predictions. We can trivially convert samples from the residual to full samples by simply adding the mean prediction to the samples.

Given an image I , we first feed it through the frozen Sam3DBody model to obtain parameter predictions Θ . We train a LoRA adapter on the promptable decoder of the backbone to output the flow context c . We concatenate the backbone predictions to the flow context and project them to obtain the full context, which is fed into the normalising flow model to output parameter residual samples $\Delta\Theta$.

v1 joint model, Figure 3. Since the model is single-view, the distribution it learns can only condition on one image $p(\theta_i, \beta_i | I_i)$. The first iteration of our model models exactly this.

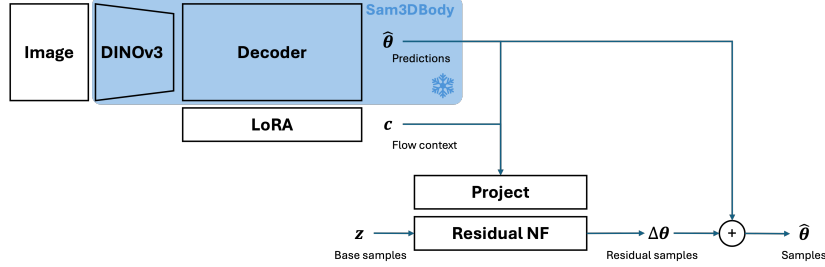


Figure 3: Joint model.

However, this does not support sampling from the marginal joint posterior, as we cannot evaluate the marginal probability $p(\beta_i | I_i)$ of the samples, even though they are properly distributed according to the marginal distribution due to the automatic marginalisation of Monte-Carlo samples.

v2 autoregressive model, Figure 4. We therefore factorise this into a two-stage network.

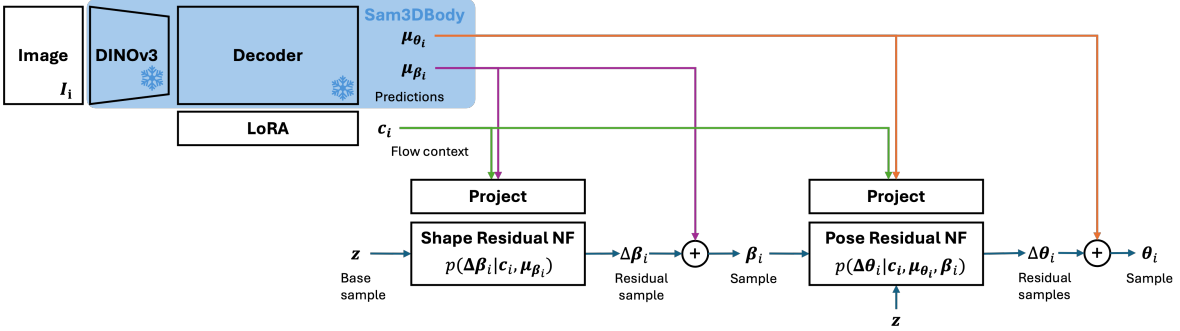


Figure 4: Autoregressive model.

The first stage learns the marginal conditional distribution $p(\Delta\beta_i | c_i, \mu_{\beta_i})$, where $c_i = f(I_i)$ is the flow context. We also explicitly concatenate the mean predictions from the frozen model $\mu_{\beta_i} = S(I_i)$ to the context, where S is Sam3dBody. *Intuitively this is: What are the plausible shape changes given the image and the mean prediction?*

The second stage learns $p(\Delta\theta_i | c_i, \mu_{\theta_i}, \Delta\beta_i)$, where $\Delta\beta_i \sim p(\Delta\beta_i | c_i, \mu_{\beta_i})$, where $\beta_i = \Delta\beta_i + \mu_{\beta_i}$ is a sample from the first stage. *Intuitively this is: What are the plausible poses given a shape sample?*

Justifying the necessity of the second stage. Our end goal is to learn the first stage network. this already allows us to achieve the end goal. So is modelling pose necessary at all? It might make it easier, as model learns using reprojection losses. Given a shape sample, the pose that satisfies 2D consistency is no longer the mean predicted pose. If we just use the mean, then the reprojection consistency manifold is collapsing, and model doesn't learn anything.

2.3.3 Losses

2D reprojection loss. We use a 2D reprojection loss on the visible samples.

$$\mathcal{L}_{2D} = \|w_i(\Pi(\mathcal{JM}(\Theta), c) - J_{gt})\|_2^2 \quad (22)$$

Parameter loss. We wish the ground truth parameters have high likelihood under the learned flows.

$$\mathcal{L}_{\Theta} = p(\Theta_{gt}|I) \quad (23)$$

3 Experiments

3.1 Implementation details

The Sam3DBody MHR parameter head is very light-weight. As a result, the frozen features from the encoders is highly specialised. An additional MLP head struggles to learn meaningful uncertainties from it. Test the following solutions:

1. Use the same MHR token - doesn't work
2. Attach an additional uncertainty token to the decoder - doesn't work
3. Use the LoRA adapter - training shape uncertainty only, doesn't work and shouldn't work (?). On NF model seems to be working.

3.2 Results

Shape only - doesn't work. Without pose uncertainty modelling, when 2D consistency constraint is present, samples should degenerate.

Diagonal Gaussians - doesn't work, We've argued that this is insufficient.

Full-Covariance Per-joint Gaussians - doesn't work, We've argued that this is insufficient - only per-joint, manifold is complex

Normalising Flows

3.3 Datasets

BEDLAM, SSP-3D, 4D-Dress

References

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